

Meeting Brief: Fuzzy Audiogram Project

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Monday 27 July 2026, 11:00 PM Riyadh (UTC+3)

1. Project Overview

Title: A Fuzzy Logic Framework for Sensorineural Hearing Loss Classification

What it does: A Mamdani-type Fuzzy Inference System (FIS) that maps pure-tone audiogram thresholds into a continuous **Fuzzy Audiometric Index (FAI)** — preserving the gradation that conventional crisp WHO thresholds discard.

Data source: NHANES 2017–2020 (n = 5,147 participants, 80/20 train/test split)

Status: Pre-submission — manuscript drafted, code open-sourced, validation complete on NHANES held-out test set.

2. Key Results

Metric	Value
FIS Rule Base	48 Mamdani-type rules (4 groups)
Weighted vs. PTA-4	0.89 (excellent agreement)
Borderline accuracy (± 5 dB)	91.2% — vs. 68.4% for PTA-4 ($p < 0.01$)
Configuration classification	89.4% accuracy (6 shapes, $\kappa = 0.86$)
Sub-threshold progression detected	28% of longitudinal cases
Spearman vs. PTA-4	0.96

Clinical advantage: At the 25/26 dB boundary, the FIS assigns graded membership (Normal = 0.40, Mild = 1.00) rather than a forced binary label.

3. Current Assets

Asset	Location
Source code & pipeline	github.com/ameye/fuzzy-audiogram
Manuscript (PDF/DOCX)	Ready for submission
Comprehensive technical report	12 sections, all analyses documented
All figures	7 manuscript + 9 EDA figures
NHANES classification results	4,475 participants classified (CSV)

4. Proposed UCH Ibadan Collaboration

What UCH can provide:

- **Clinical audiogram dataset** (500+ ears) — real-world Nigerian population with diverse configurations
- **ENT expertise** — rule base refinement, clinical interpretation
- **Validation cohort** — Menière's, otosclerosis, CSOM patterns underrepresented in NHANES

What we offer in return:

- Co-authorship on manuscript (validation section)
- Joint development of web-based FAI Calculator for clinical use
- CV-based audiogram extraction tool (currently in planning)
- Open-source credit and academic collaboration

Timeline:

- Phase 1 (Aug 2026): Collect audiograms → run through existing FIS → compare with ENT clinical grading
 - Phase 2 (Sept 2026): Publish validation results — submit manuscript to *Ear & Hearing* or *Int J Audiology*
 - Phase 3 (Oct 2026): Deploy FAI Calculator + CV scanner at UCH (pilot)
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5. Talking Points

1. **“The FAI doesn’t replace clinical judgment — it augments it.”** The system outputs interpretable membership degrees, not black-box probabilities. Every rule can be inspected.
 2. **“The borderline problem is real — 39% of NHANES ears fall within ± 5 dB of a WHO boundary.”** This is where the fuzzy system adds the most value.
 3. **“UCH data would strengthen the manuscript significantly.”** The current study uses only NHANES (US population). Nigerian clinical data would demonstrate generalisability across populations and healthcare settings.
 4. **“We can build an FAI Calculator together.”** A web-based tool where ENT surgeons can enter thresholds and get instant FAI scores + configuration vectors + longitudinal tracking.
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6. Links

- **GitHub:** <https://github.com/ameye/fuzzy-audiogram>
 - **Comprehensive report:** [comprehensive_report.pdf](#) (3.8 MB)
 - **Manuscript:** [manuscript.pdf](#) (2.0 MB)
 - **Contact:** sanyaameye@hotmail.com
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Brief prepared by Hermes Agent — 27 July 2026